Pneumatic Gun



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Pneumatic gun

What it is used for

A pneumatic gun uses compressed air to perform tasks such as blowing debris, applying sealant, fastening, or other operations depending on the gun type. Use the correct accessory and pressure rating, and follow the tool-specific operating instructions.

Main parts

- Gun body and grip
- Trigger or operating lever
- Air inlet and quick-connect fitting
- Nozzle, barrel, or task-specific attachment
- Pressure regulator or adjustment control, if fitted
- Safety lock and protective accessories, if fitted

Before you start

1. Identify the pneumatic gun type and confirm its intended application.
2. Inspect the body, trigger, nozzle, air hose, and connectors for damage.
3. Confirm the compressor and hose pressure ratings match the tool requirements.
4. Check that couplings are securely fitted and the air supply is regulated correctly.
5. Fit the correct nozzle or attachment and inspect it for blockage or wear.
6. Wear appropriate PPE, including eye protection and hearing protection where required.

How to use

1. Disconnect the air supply before fitting accessories or inspecting the gun.
2. Secure the approved nozzle or attachment according to the manufacturer's instructions.
3. Connect the gun to the correct regulated compressed-air line.
4. Set the air pressure within the tool's specified operating range.
5. Check the connection for leakage using an approved method.
6. Hold the gun securely and keep the nozzle directed toward the work area.
7. Operate the trigger using the technique specified for the tool's purpose.
8. Maintain a safe distance from people and sensitive equipment.
9. Monitor the tool, hose, and workpiece for abnormal vibration, leakage, or malfunction.
10. Release the trigger and disconnect the air supply before clearing a blockage or changing settings that require isolation.
11. After use, shut off the air supply and release residual pressure according to the procedure.
12. Disconnect the hose safely and store the tool and accessories in a dry place.

⚠ Safety

- Only trained users should operate the pneumatic gun for its intended task.
- Never direct compressed air at people, skin, or clothing.
- Do not exceed the tool, hose, or accessory pressure rating.
- Never use damaged hoses, loose couplings, or a faulty trigger.
- Depressurise and disconnect the air supply before maintenance or clearing blockages.
- Use task-specific PPE and follow additional safety requirements for fastening or impact-type guns.

Care & storage

- Clean the nozzle and external surfaces after use.
- Inspect the hose, couplings, trigger, and attachments regularly.
- Lubricate only if required by the manufacturer and use the specified lubricant.
- Store the gun disconnected from compressed air in a dry location.
- Remove faulty tools from service and arrange repair by authorised personnel.

Common mistakes to avoid

- Setting air pressure above the tool's rated limit.
- Directing compressed air at skin or another person.
- Changing attachments while the air supply remains connected and pressurised.
- Ignoring air leaks, damaged hoses, or a sticking trigger.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar